



Research paper

Dynamic volatility spillover between oil and marine shipping industry

Adeel Riaz^a, Li Xingong^a, Zhilun Jiao^{b,c,*}, Muhammad Shahbaz^d^a Business School, Henan University, Henan, China^b College of Economic and Social Development, Nankai University, Tianjin, China^c The Laboratory of Behavioral Economics and Policy Simulation, Nankai University, Tianjin, China^d School of Management and Economics, Beijing Institute of Technology, Beijing 100081, China

ARTICLE INFO

Article history:

Received 1 August 2021

Received in revised form 6 February 2023

Accepted 10 February 2023

Available online 24 February 2023

Keywords:

Oil volatility

Volatility spillovers

Marine shipping industry

Oil price risk

ABSTRACT

Oil consumption not only makes up a large percentage of the overall operating expenses for the marine shipping industry, besides that, the tanker sector is a major carrier of global oil supply, which magnifies the relevance of oil market for the shipping industry. In this backdrop, the following study examines the volatility transmission between marine shipping industry's tanker and dry cargo market and oil market using daily data from May 2006 to August 2021 by employing spillover index methodology. The empirical results suggest pronounced spillovers for the tanker market while lower spillovers are observed for dry cargo market, indicating the well-integrated tanker market in comparison to dry cargo sector and oil market. The volatility in oil prices contributed higher spillovers in the tanker market, and remained a net contributor. The volatility in oil prices is a significant source of the volatility transmission during periods of the sudden drop in oil prices coupled with the higher volatility in oil market, and is a net receiver during stable periods. The period of higher spillovers lasts longer for the tanker sector as compared to dry cargo sector. In the particular case of dry cargo market, the smaller (larger) the size of the vessels, the greater (lower) are the spillovers observed (From and To). Hedging turns out to be irrelevant during turmoil periods, as the comovement between shipping and oil market becomes stronger. The spillovers are pronounced during the period of financial crisis, COVID-19 and 2014–2016, owing to the significant fluctuations in oil prices and a troubled period for marine shipping industry.

© 2023 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

1. Introduction

Oceans are the highways for the economic development and had been used for transportation for centuries. Smith (1776) described shipping as an economical way of transporting goods and regarded shipping as a catalyst for the economic development. Seaports around the world handle about 80 percent of the global trade by volume, and most (70%) of it is carried on by marine shipping (Unctad, 2017). With the evolution in the shipping industry, the challenges, risks and benefits also altered with the time. Shipping is no different than other industries in that it depends on fuel and is affected by the volatility in oil prices. An industry's sensitivity to the volatility in oil prices depends on the relative proportion of oil as an input; its ability to pass this volatility in oil prices to its customers and hedging. Fuel cost is the largest cost item of a ship's operational expenses

commonly referred to as OPEX. Sometimes it reaches to almost 50–70 percent of the voyage cost (Chou et al., 2017; Lau et al., 2013). According to Unctad (2017), fuel cost is the key cost while transporting goods by marine shipping. An upsurge in oil prices during 1980's resulted in laying up ships with high fuel consumption. Thereafter, shipping industry focused on the design of the ship in order to reduce fuel consumption. After the 1980's, the biggest shock to the shipping industry was the financial crisis of 2008. During the financial crisis, several shipping companies either merged or went bankrupt because of the lower global demand. Although shipping industry depends heavily on fuel prices yet the increase in its revenue during the periods of higher fuel prices before the financial crisis is thought-provoking. On the other hand, oil prices dropped significantly after the financial crisis of 2008, which turned out to be glad tidings for the shipping industry in the form of lower operating expenses. But it is interesting to note that shipping industry also endured its worst times during the same period.

Fuel consumption is a major concern for shipping companies during a voyage. Ship-owners face a dilemma in the form of trade-off between speed and fuel consumption. Fuel consumption increases the operating cost for shipping while a slower

* Corresponding author at: College of Economic and Social Development, Nankai University, Tianjin, China.

E-mail addresses: adeelriaz@henu.edu.cn (A. Riaz), lxg@henu.edu.cn (L. Xingong), zjiao@nankai.edu.cn (Z. Jiao), muhdshahbaz77@gmail.com (M. Shahbaz).

speed yields lower revenues. Numerous researchers have tried to investigate this relationship to make sails more economical (Bialystocki and Konovessis, 2016; Fagerholt et al., 2010; Psaraftis and Kontovas, 2014; Ronen, 1982; Wang et al., 2013). Ship-owners are so much concerned about the fuel consumption of a ship that even weather factors like wind speed are considered in order to save fuel cost (Bialystocki and Konovessis, 2016). A higher speed not only increases the cost for shipping in terms of fuel consumption but it also results in higher pollution thereby increasing carbon emissions; which is also becoming a major concern for shipping industry due to the stringent environmental regulations. As a result, higher revenues come at the cost of higher fuel consumption coupled with higher carbon tax. Another way shipping industry is affected by increase in oil prices is inflation. Shipping industry relies on global demand of goods and a higher inflation results in lower global demand because of the increase in the oil price. Shipping industry is a capital-intensive industry and increase in inflation consequently increases interest rate thereby increasing financial costs for shipping. The tradeoff between fuel saving and speed, inflation, interest rates and lower demand emanates from oil prices and makes shipping industry highly vulnerable to the volatility in the fuel prices.

This study delivers a useful insight to the stakeholders of shipping industry about the contagion effects of oil prices to the marine shipping industry, which will aid in the better comprehension of the relationship between the fluctuations in the oil market and the marine shipping industry. Furthermore, the analysis in the current study will also highlight the sector (dry cargo vs tanker) of the shipping industry which is more exposed to the volatility in the oil prices. It could also help in understanding the question of whether an industry (tanker sector) which derives its demand from oil, and is also a consumer of oil, is more exposed to the fluctuations in oil prices or remains invulnerable. The contributions of our research are fivefold. First, previous studies (Chen et al., 2010; Dai et al., 2014; Drobetz et al., 2012; Kavussanos, 2003; Kavussanos and Marcoulis, 1997; Tsouknidis, 2016) have focused on the dynamics of volatility spillovers among different shipping sectors; however, the spillovers from a highly relevant external factor i.e. oil price's fluctuations has largely been ignored. Second, the following study complements the literature concerning the dynamic spillovers by incorporating the volatility in the oil prices. Although recently, Maitra et al. (2020) observed the spillovers from the volatility in oil prices; however, their research was focused on twelve liner shipping companies while disregarding the tanker sector, the latter being a main consumer and carrier of oil. Third, this study not only incorporates the tanker and dry bulk market but also considers the sub-divisions of these sectors, which gives a useful picture of how different sub-sectors of the shipping industry are interconnected among themselves and with the volatility in oil prices. Fourth, the following study not only considers the implications of financial crisis but also includes the period of COVID-19 and compares both periods in order to have a deep understanding of how the volatility in oil and different sectors interacted during these highly volatile periods. Fifth, the following study also suggests optimal hedging ratios and optimal portfolio for better risk management. Investors are highly concerned about the fluctuations in market responsible for downside risk. The volatility in oil market can pose a greater risk, and an understanding of the transmission of this risk will be helpful in identifying and hedging oil prices risk in the shipping industry thereby guarding the risk averse investors from the probable losses.

The next section of the paper provides a brief overview of the literature about how the fluctuations in the oil prices affect industries and why it matters for the marine shipping industry. The third section discusses about the data and methodology. Section 4 details results and discussion followed by the robustness of our analysis. The last section concludes the main findings and policy implications of our study.

2. Literature review

Earlier studies relating to the effects of fluctuations in oil prices were primarily fixated on economies. Thereafter, scholars shifted their attention from economies to investigate the impact of volatile oil prices on various oil dependent industries. For example, Hamilton (1983) showed that oil price shocks have adverse impacts on global economy to such an extent that it can lead to a recession. An appreciation in oil prices affects economies by inflating the prices of goods and increasing interest rates, thereby limiting access to cheap capital for industries. Researchers have noted a significant impact (positive and negative) of changes in oil prices on economies (Akinsola and Odhiambo, 2020; Anton Mork et al., 1994; Aydin and Acar, 2011; Kilian and Vigfusson, 2011; Lee et al., 1995) and stock market (Aggarwal et al., 2012; Alamgir and Amin, 2021; Al-hajj et al., 2018; Basher et al., 2012; Faff and Brailsford, 1999; Jones and Kaul, 1996; Kocaarslan and Soytas, 2019; Mokni, 2020; Park and Ratti, 2008; Sadorsky, 1999; Soytas et al., 2009; Tiwari et al., 2018; Tuna et al., 2021). The relationship between oil prices and commodity markets has also been documented in the literature as the fluctuations in the oil price influence gold, silver and agriculture commodity future markets (Junttila et al., 2018; Kang et al., 2017; Li et al., 2021; Soytas et al., 2009; Tiwari et al., 2020; Tiwari and Sahadudheen, 2015). Moreover, the effects of the oil prices have also been observed for the forex markets (Basher et al., 2012; Cifarelli and Paladino, 2010; Huang and Li, 2022; Saidu et al., 2021; Xu et al., 2019). Although the co-movement between oil prices and other markets and economies had been widely studied; however, recently, scholars have started investigating the time varying spillover effects of the oil prices across economies and industries specifically after the financial crisis.

The volatility in oil prices is a long-standing characteristic of world oil markets. Spillovers from oil and other commodities drew attention of researchers after the drastic effects of financial crisis across the globe. As the crisis intensified so did the spillovers across markets started to increase (Diebold and Yilmaz, 2012). However, diverse results have been reported for oil's volatility spillovers across different sectors of an economy (Cong et al., 2008; Diaz et al., 2016; Park and Ratti, 2008). The transmission of volatility spillovers from oil to stock market (Malik and Hammoudeh, 2007; Sadorsky, 2012; Singhal and Ghosh, 2016; Soytas and Oran, 2011), and commodity market (Choi and Hammoudeh, 2010) has been well established among the researchers. The spillovers from the volatility in oil prices have also been observed in the context of markets such as food (Du et al., 2011; Fasanya and Akinbowale, 2019; Nazlioglu et al., 2013), forex (Singh et al., 2018) and agricultural and non-agricultural commodities (Kang et al., 2017; Luo and Ji, 2018). Despite the time varying spillovers between oil prices and industries which are not a direct consumer of oil; the spillovers from oil to the marine shipping industry have largely been ignored. Marine shipping industry is not only a huge consumer of oil but happens to be the major source of transportation of oil and is sensitive to the uncertain events in the oil market and to the major economic events (Fei et al., 2020). Although, Sun et al. (2019) observed the effect of dynamic spillover from oil futures market for the tanker sector of shipping market. However, they fail to account for the bulk sector of shipping industry, which constitutes a major pie of shipping market and is a huge consumer of oil. Fuel cost is a major component of the operating cost of bulk shipping also. In another study about shipping, Tsouknidis (2016) investigated the dynamic volatility spillovers among the different sectors of marine shipping, but the spillovers from other markets e.g. oil market, commodity markets etc. have not been sought after. However, given the strong dependence of the marine shipping

industry on oil, the spillovers from oil market need further attention for those sectors also which had hardly been investigated for risk transmission from oil and are massive consumers of oil.

The impact of a change in the oil price is heterogeneous in respect of different industries. Realizing the importance of this heterogeneous impact of oil prices, researchers have attempted to study how oil prices affect different industries by learning its impact across industries. Peng et al. (2018) showed that the effect of crude oil price is heterogeneous across industries and a positive relationship exist between oil price and market returns specifically during the bearish market. They were also of the view that industry features are an important consideration while investing the impact of oil price volatility. By using over twenty years of data, Aggarwal et al. (2012) found negative association between oil price shocks and the transportation sector firms' returns. Kristjanpoller and Concha (2016) analyzed the impact of oil price fluctuations on airline industry using CAPM and GARCH model. They found positive relationship between oil price fluctuations and airline's stock on daily basis. Mohanty et al. (2014) employed Fama–French and Carhart model in case of U.S. travel and leisure industry and concluded that the sensitivity of the oil prices varies significantly across sub-sectors. They also showed that oil price's exposure is time varying and oil price risk exposure was significantly increased for the airline industry during the financial crises of 2008. Scholtens and Yurtsever (2012) also studied the relationship at an industrial level in the Euro area and found that the response of different industries to oil price shocks differs across industries. Oil price shocks also affect industries beyond direct input cost by affecting the demand and supply in certain industries. In case of automobile industries these shocks reduces demand but for industries where the share of the oil is larger as an item of cost it is responsible for reducing supply (Lee and Ni, 2002). This affect is not limited to demand and supply of industrial production, the volatility in oil price can affect industries in disparate ways. An unanticipated rise in price can have positive, negative or insignificant implications for the industries. A positive relationship is testified in case of European oil and gas sector (El-Sharif et al., 2005). They also reported a weak relationship between oil and returns of non-oil and gas sectors. Ramos and Veiga (2011) investigated how oil price affects oil and gas industry in 34 countries and found that the oil and gas sector in the developed countries responds more strongly to changes in the oil price and the relationship is not strong in case of emerging economies. He also observed asymmetric effect of oil price and showed that a rise in oil price impacts more severe than a decline in oil price. Narayan and Sharma (2011) examined the relationship between firm's returns and oil price for US and found that oil price affects firms' returns differently depending on their size and he further found that oil price and transportation sector is positively related. However, this positive effect is observed for only 31% of the transportation sector. Since the transportation sector is composed of ground transport, sea transport and air transport so the applicability of the finding to a sector is uncertain. So, this study fills this gap for shipping which is an important part of the transport sector. Shipping industry, an important part of transportation sector, had not been studied in isolation although it is heavily dependent on oil as input. Moreover, oil constitutes a major operating expense for the marine shipping industry.

Fuel makes up about 50% of a ship's voyage cost (Stopford, 2009). Because of lower oil prices before the 1970's, fuel cost made up about 13% of the total shipping costs. However, a sharp upsurge in oil prices during the 1980's augmented fuel cost to 34% of the total shipping costs. Majority of the researchers investigated the impact of oil prices from the lens of cutting the cost during the voyage; however, the impact of the change in the oil

prices on the overall shipping markets had been largely ignored. Ronen (1982) was among the earliest researchers who examined the tradeoff between speed and fuel consumption in the case of marine shipping. He suggested three models for the optimal speed of ships in order to reduce loss in revenues owing to the fuel consumption. Yao et al. (2012) developed a fuel management strategy to reduce fuel consumption during a voyage and was of the view that better operational strategies can help in reducing fuel cost for shipping. While calculating items of revenues for shipping, Veenstra and Ludema (2006) included expected fuel consumption since a lower fuel costs result in higher cash flows. Chou et al. (2017) studied the impact of navigation efficiency for transiting among different ports. They derived navigation efficiency from fuel consumption. Bialystocki and Konovessis (2016) suggested an operational approach for fuel consumption and speed curve and considered weather factors apart from other factors. They quantified weather factors like wind speed and revealed that these factors significantly contributed in fuel savings. Psaraftis and Kontovas (2014) addressed issues related to speed optimization and incorporated various considerations of shipowners. Wang and Meng (2012) used operating data to examine the optimal speed in case of container ship industry. The relationship between oil consumption and speed of a ship is exponential in nature (Veenstra and Ludema, 2006). Reduction in speed by 20% cuts fuel expenses by 50% (Ronen, 1982). Fagerholt et al. (2010) stated that fuel consumption of a ship on a particular route is a cubic function of speed and suggested alternative routes for ships in order to save fuel consumption. A higher speed burns more fuel and produces more carbon dioxide for which shipowners have to pay additional carbon tax (Chou et al., 2017). Choice of fuel impacts operating costs in terms of carbon tax. Fuels which emit a higher level of carbon in the air are being replaced with the ones resulting in lower emissions. Abadie et al. (2017) suggested techniques to have a balance between fuel and carbon emissions by changing fuel mix, installing scrubber and operating in Emission Control Area (ECA).

Scholars have emphasized that industry is an important consideration while studying the impact of oil prices. The risk transmission from oil depends on industry characteristics (Peng et al., 2018), the extent to which the particular industry is dependent on oil (Phan et al., 2015), and varies across industries (Hedi Aroui and Khuong Nguyen, 2010; Sadorsky, 2008; Yun and Yoon, 2019). The risk-return profile of shipping industry is different from other industries. Therefore, it demands that the risks related to marine shipping industry be investigated separately. Drobetz et al. (2010) suggested that risk and returns of shipping industry are different from other industry indices so it has the potential to serve as a separate asset class. Therefore, following study focuses on the risk transmission from volatile oil market to shipping industry, which is not only dependent on but is also the major carrier of oil, and according to researchers should be treated as a separate asset class.

3. The data and econometric model

3.1. The data

Our analysis is based on the daily volatility in oil price, dry cargo and tanker sectors of marine shipping industry for the period starting from May 2006 to August 2021. During the chosen timeframe, shipping industry had not only experienced a boom period but had also faced the worst economic conditions. Moreover, a period of boom coupled with a period of crises for shipping firms can give a better picture of how volatility spillovers of oil prices behaved during the volatile times. The period is also significant as it not only includes the financial crisis of 2008

Table 1
Descriptive statistics.

	BRENT	BCTI	BDTI	BCI	BPI	BSI	BHSI
Mean	0.0013	−0.0214	−0.0195	−0.0195	0.0138	0.0147	0.0172
Median	0.0395	−0.1750	−0.1419	−0.1682	0.0000	0.0335	0.0000
Maximum	41.2023	29.2244	22.9511	162.9241	23.5498	20.2794	9.8147
Minimum	−64.3699	−29.6472	−38.1224	−453.2599	−21.6235	−11.6631	−13.7681
Std. Dev.	2.8254	2.1423	2.2932	10.7892	2.8327	1.6345	1.2903
Skewness	−2.4085	1.6500	−0.4376	−18.6281	0.0995	0.2744	−0.9374
Kurtosis	99.3310	52.8684	43.5377	870.8670	9.2566	16.2531	13.8092
Jarque–Bera	1442332	387255	254900	117000000	6075	27279	18660
{p-value}	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
ADF(Unit root)	−14.5379*	−25.6507*	−26.1497*	−24.2275*	−22.0995*	−15.0099*	−12.8928*
{p-value}	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)

Note: Unit root test is conducted at level with an intercept term by using (Dickey and Fuller, 1981) and is denoted by ADF. Lag length is determined based on minimizing the SIC (Schwarz information criteria).

*Denotes significance at 1%.

but it also takes into account the COVID-19 period which carries significant implications for the economies around the world. Due to the unavailability of the data for the Baltic Exchange Supramax Index (BSI) and Baltic Exchange Handysize Index (BHSI) before 2006, the period of our study starts from 2006. For the dry cargo market, we consider Baltic Handysize index (BHSI), Baltic Supramax index (BSI), Baltic Capesize index (BCI) and Baltic Panamax index (BPI). To investigate the effects of oil spillovers for the tanker market of the marine shipping industry we consider Baltic Dirty Tanker Index (BDTI) and Baltic Clean Tanker Index (BCI). The different sectors and sub sectors are selected since these sectors have unique business cycles (Kavussanos and Visvikis, 2016) in comparison to each other, and they also exhibit segmentation based on vessel size and type (Kavussanos, 2003). The shipping tanker market derives its demand from oil (Drobtz et al., 2012), therefore, it is closely affiliated with the oil market. It is not only the major carrier of global oil but also uses oil as fuel. For oil prices, we use daily Brent spot prices obtained from Energy Information Administration (EIA) in US dollars. Brent oil prices are widely used in the industry for hedging, futures and forwards (Aggarwal et al., 2012), and are one of the major benchmark for oil prices. We use spot prices instead of futures prices since most of oil prices shocks in the literature are studied based on the spot prices (Aggarwal et al., 2012). The data were obtained from DataStream which is a widely used and trusted database among financial researchers for global equity indices and equity returns for different countries and across different industries. The volatility for the returns is computed from the conditional variances generated from GARCH (1, 1) process (Kindly see the Appendix-1 for volatility computed from DCC GARCH). The daily change in the indices ($C_{i,t}$) is computed using the following expression:

$$C_{i,t} = 100 \times \ln(PRICE_{i,t}/PRICE_{i,t-1})$$

where $PRICE_{i,t}$ is the value of an index i at time t while $PRICE_{i,t-1}$ will be value of an index i for the previous period i.e. $t - 1$.

We present the descriptive statistics for all the returns series in Table 1. The mean returns are negative for the tanker market and BCI while the returns for the oil and dry cargo market are positive with the exception of BCI which also exhibits higher standard deviation. Moreover, we also observe maximum returns for the Baltic Capesize index, which is quite high in comparison to other series for which maximum varies between 20 to 42. The difference between maximum and minimum values can also be seen in the form of higher standard deviation for Capesize vessels. Lower volatility indicated by standard deviation is observed for smaller vessels (BSI and BHSI). Most of the markets exhibit positive skewness with the exception of BCI, oil and dirty tanker stipulating the presence of frequent small losses with large gains. While examining the descriptive statistics for the tanker sector, we note higher kurtosis which indicates the presence of higher

number of outliers. Furthermore, higher kurtosis for all the series in our dataset also demonstrates that distributions of returns are relatively fat-tailed and peaked and the same is true in the case in dry cargo sector. All the return series are stationary at level (included intercept term) by using (Dickey and Fuller, 1981) test. The null of normality is strongly rejected for all the series as indicated by Jarque–Bera normality test.

3.2. Econometric model

To measure the volatility spillovers across different sectors of the marine shipping industry and the volatility in the oil prices, we employed the connectedness measures model suggested by Diebold and Yilmaz (2012). The model is based on a covariance stationary N-variable VAR model with P lag length. Let volatilities be $\hat{E}_i$ modeled as a VAR(p) taking the form:

$$\hat{E}_i = \sum_{i=1}^p \Pi \hat{E}_{t-i} + \epsilon_t \quad (1)$$

where

$$\hat{E}_t = (\hat{E}_{1,t}, \hat{E}_{2,t}, \hat{E}_{3,t}, \dots, \hat{E}_{N,t})$$

and Π is an $N \times N$ matrix of the parameters to be estimated. ϵ is the vector of error term which is independently and identically distributed with zero mean and covariance matrix. The transformations of the formula into moving average of the Eq. (1) takes the form:

$$\hat{E}_t = \sum_{i=0}^{\infty} \Lambda \epsilon_{t-i} \quad (2)$$

and $N \times N$ co efficient Λ obeys a recursion of the form

$$\Lambda_i = \Pi_1 \Lambda_{i-1} + \Pi_2 \Lambda_{i-2} + \dots + \Pi_p \Lambda_{i-p} \quad (3)$$

The initially introduced method by Diebold and Yilmaz (2009) has methodological limitations e.g. dependence of resulting variable decompositions on variable orderings since the framework was based on Cholesky factor orthogonalization within a VAR estimation framework. The discrepancy regarding ordering was resolved by replacing it with framework for generalized VAR (Koop et al., 1996; Pesaran and Shin, 1998). The method was more useful since it also generated directional spillovers, which gives a clear picture of each variable in the model and contributes in the overall model. Based on Eq. (1), we decompose H-step-ahead forecast error variance and analyze the forecast error of variable ξ_i due to shocks of variable ξ_j for $i \neq j$. Hence $\xi_{ijg}(H)$ for (H step ahead) is computed as:

$$\xi_{ij}^g(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (e_j' h_h \sum e_j)}{\sum_{h=0}^{H-1} (e_j' h_h \sum e_j)} \quad (4)$$

where Σ is the variance matrix of the vector of errors ϵ , and for the j th shipping sector σ_{jj} is standard deviation of the error term. Finally, e_i is a selection vector taking value of 1 on i th element, and zero otherwise. For the shocks to each variable that are not orthogonalized, we have to normalize the variance decomposition shares in order to ensure that each array of the spillover matrix is equal to one. However, when applying the generalized VAR modeling, each row in the table of variance decomposition is not equal to one indicating that the sum of portions for own and cross variable variance contributions is not equal to one. In order to obtain a unit sum of each row of the variance decomposition, we normalize each entry of the matrix by the row sum as:

$$\hat{\xi}_{ij}^g(H) = \frac{\xi_{ij}^g(H)}{\sum_{j=1}^N \xi_{ij}^g(H)} \quad (5)$$

Where

$$\sum_{j=1}^N \xi_{ij}^g(H) = 1$$

and

$$\sum_{j=1}^N \hat{\xi}_{ij}^g(H) = N$$

Total spillover index which measures the contributions of all of the variables' shocks "to" and "from" the forecast errors is calculated as:

$$Sp^g(H) = \frac{\sum_{i,j=1}^N \hat{\xi}_{ij}^g(H)}{\sum_{i,j=1}^N \hat{\xi}_{ij}^g(H)} \times 100 = \frac{\sum_{i,j=1}^N \hat{\xi}_{ij}^g(H)}{N} \times 100 \quad (6)$$

It calculates the extent of spillovers among the shipping sectors i.e., BDI, BDTI and oil and also for the dry cargo sector among BCI, BPI, BSI and BHSI. Gross spillovers are the sum of proportions of the forecast error variance of $\hat{\epsilon}_i$ due to shocks to $\hat{\epsilon}_j$ for all $i \neq j$. Diebold and Yilmaz (2012) model is advantageous since it also aids in calculating the directional spillovers. The volatility spillovers transmitted from oil (Brent) to the shipping market i.e., the dry cargo and the tanker sectors is computed as:

$$Sp_{i \rightarrow j}^g(H) = \frac{\sum_{i,j=1}^N \hat{\xi}_{ij}^g(H)}{\sum_{j=1}^N \hat{\xi}_{ij}^g(H)} \times 100 \quad (7)$$

Although oil is generally considered to be a major transmitter of spillover, it also receives volatility spillovers from other markets. The directional spillovers received by the oil market from the shipping sectors are calculated as:

$$Sp_{j \rightarrow i}^g(H) = \frac{\sum_{i,j=1}^N \hat{\xi}_{ji}^g(H)}{\sum_{j=1}^N \hat{\xi}_{ji}^g(H)} \times 100 \quad (8)$$

The net spillovers assess whether a particular asset or market is the net receiver or transmitter of the return or volatility spillovers effect and is the difference between Eqs. (6) and (7) computed as:

$$Sp_i^g(H) = Sp_{i \rightarrow j}^g(H) - Sp_{j \rightarrow i}^g(H) \quad (9)$$

4. Results and discussion

We present the volatility spillover index matrix for the tanker and dry cargo sectors in Tables 2 and 3. The empirical findings indicate a higher overall volatility spillover in case of tanker sector (31.20%) in comparison to the dry cargo sector (13.60%). The higher overall spillovers among the tanker market show that tanker market is well integrated in comparison to dry cargo

sector. The results for volatility spillovers are based on VAR of order 2 for tanker and order 1 (based on Schwarz information criteria) for dry cargo with 10-week-ahead forecast errors for generalized variance decomposition. The volatility in oil prices plays a significant role in transmitting and receiving spillovers in the tanker sector. The contribution of oil volatility spillovers is much more pronounced and amounts to 26.4% as regards tanker market. On the receiving end too, oil volatility receives up to 13.3% from clean and dirty tanker sectors. In case of directional spillovers, clean tanker transmits the most spillovers of about 42% but it receives 47% of spillovers from all others. On net, BDTI receives more volatility spillovers than clean tanker sector while oil transmits about 13%. BCTI is followed by oil which transmits about 26% volatility, and it receives 13% from clean and dirty tanker sector. Overall, we observe higher spillovers for tanker market in comparison to dry cargo market, and the directional and net spillovers discussed in the next section will give us a complete picture of which indices are responsible for the higher spillovers in our analysis.

Higher spillovers in the case of tanker market are observed owing to the greater dependence of tanker sector on the oil market. Moreover, the directional spillovers also indicate the greater integration of tanker sub-sectors. Tanker shipping is responsible for the carriage of oil from port to port and also consumes a larger amount of oil as input. In the case of tanker sector, the biggest contributor of the spillovers is the clean tanker market and it also receives the most from others followed by the dirty tanker sector. The oil volatility receives spillovers amounting to 13.3% while it contributes about 26.4% as regards tanker market but its influence on cargo market remains insignificant. In the case of bidirectional spillover, BCTI and BDTI have more effect on each other than the volatility spillovers from oil. A somewhat similar findings were noticed by Tsouknidis (2016), who observed the volatility spillovers from the Baltic Dry Index (BDI) to tanker market and found greater volatility spillovers between clean tanker and dirty tanker rather than "from" or "to" dry cargo sector. As regards dry cargo sector, smaller vessels such as Handysize and Supramax are the biggest contributors and receivers of spillovers while the bigger vessel index (Capesize index) remained fairly resilient to spillovers from all other dry cargo sector and oil. The results indicate an interesting phenomenon; with the increase (decrease) in the size of the vessel there is a decrease (increase) in volatility spillovers. A similar relationship was observed for smaller vessels by Tsouknidis (2016) while observing the spillovers among the shipping freight markets. Whereas, observing the dynamics of conditional volatilities, Kavussanos (1997) found that the nature of the volatilities varies across different ship sizes. In the same way, our results also indicate that the response and the magnitude of the response to the volatility shocks varies for different ship sizes of dry cargo market. Dry cargo segment is mainly concerned with the transportation of ores, coal, grain etc. It mainly uses oil as a fuel which makes up a greater chunk of the operating expenses. Moreover, the impact of oil volatility remains negligible in case of volatility spillovers for dry cargo sectors, which indicates that tanker sector is more prone to the volatility in oil prices. However, the dry cargo market remained impervious to the volatility spillovers from oil in comparison to the tanker sector.

Fig. 1 offers a useful picture about the behavior of volatility spillovers for tanker market sub-sectors and oil volatility. We realize relatively moderate spillovers for the period before financial crisis, gradually declining and stabilizing before the financial crisis, except for the period of higher oil prices during 2006. From the onset of 2008 financial crisis and specifically after fall of Lehman brothers, we observe pronounced spillovers; the highest for our sample period. The spike at the end of 2009 is owing to

Table 2
Volatility spillovers matrix-tanker (Calculated from DY spillover index method).

	OIL	BCTI	BDTI	From Others	Net	Conclusion
OIL	86.65	11.2	2.15	13.3	13.1	Net-contributor
BCTI	23.81	53.33	22.86	46.7	−4.6	Net-recipient
BDTI	2.55	30.91	66.54	33.5	−8.5	Net-recipient
Contribution to others	26.4	42.1	25	93.5		
Contribution including own	113	95.4	91.6	31.20%		

Note: The underlying variance decomposition is based on a VAR system with two lags. The (ij) value is the estimated contribution to the variance of the 10 step ahead volatility forecast error of an index i coming from innovations to volatility of index j . “Contributions to others” is the directional volatility spillovers from oil, BCTI and BDTI to others, while “from others” indicate the directional spillovers from the shipping indices and oil to the particular index. The values are expressed in percentages.

Table 3
Volatility spillovers matrix-dry cargo (Calculated from DY spillover index method).

	OIL	BCI	BPI	BSI	BHSI	From Others	Net	Conclusion
OIL	99.47	0	0	0.23	0.3	0.5	−0.3	Net-recipient
BCI	0.1	99.83	0.06	0.01	0	0.2	−0.2	Net-recipient
BPI	0.01	0.03	91.23	5.11	3.63	8.8	2.6	Net-contributor
BSI	0.04	0.01	4.51	66.93	28.51	33.1	−9.4	Net-recipient
BHSI	0.1	0.01	6.81	18.33	74.76	25.2	7.2	Net-contributor
Contribution to others	0.2	0	11.4	23.7	32.4	67.8		
Contribution including own	99.7	99.9	102.6	90.6	107.2	13.60%		

Note: The underlying variance decomposition is based on a VAR system with two lags. The (ij) value is the estimated contribution to the variance of the 10 step ahead volatility forecast error of an index i coming from innovations to volatility of index j . “Contributions to others” is the directional volatility spillovers from oil, BCI, BPI, BSI and BHSI to others, while “from others” indicate the directional spillovers from the shipping indices and oil to the particular index. The values are expressed in percentages.

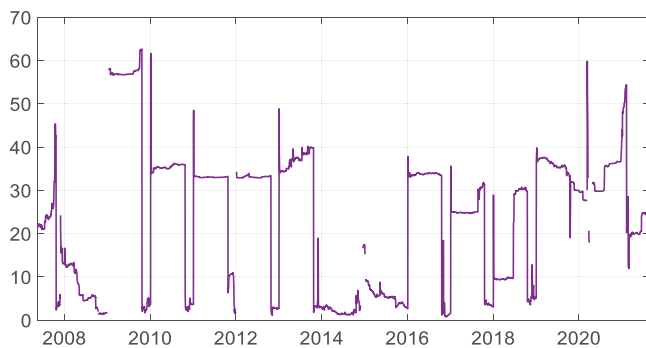


Fig. 1. Total volatility spillovers (Tanker).

Notes: Dynamic volatility spillovers are calculated from the forecast error variance decompositions on 10-step-ahead forecasts; total spillover indices are estimated using rolling window; y-axis shows percentages (%).

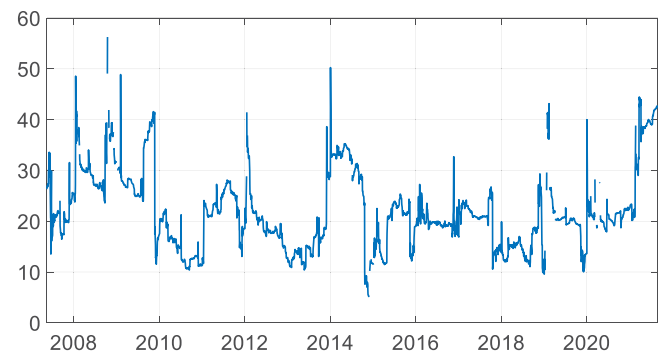


Fig. 2. Total volatility spillovers (Dry cargo).

Notes: Dynamic volatility spillovers are calculated from the forecast error variance decompositions on 10-step-ahead forecasts; total spillover indices are estimated using rolling window; y-axis shows percentages (%).

the sudden drop in tanker and other shipping sectors' indices. Another period of higher spillovers is observed from the beginning of the COVID-19. It is pertinent to note that the behavior of the peaked spillovers during the pandemic was quite similar to the period of financial crisis. However, the peak period diminished more quickly in comparison to the financial crisis as markets moved towards stability.

We observe noticeable volatility spillovers in case of tanker sector, while a slightly lower and stable spillovers can be seen for the dry cargo sector in Fig. 2. There are more periods of lower spillovers in case of dry cargo while we see sufficient number of pronounced spillovers' periods in case of tanker sector. The shipping industry was hard hit during 2011–2012 as the BDI and other shipping indices bottomed in February 2012, which is quite observable from the spike at the beginning of 2012 for both sectors. This was the period where shipping companies started to receive the delivery of the ships ordered before the financial crises. The shipping companies' orders were based on the projections of the expected world demand before the crisis. A unique feature of shipping industry is that ships are ordered well before time. The pre-crisis period proved to be remarkable for the shipping industry as it enjoyed super normal profits and industry

expected that golden period will continue for the foreseeable future. However, after financial crisis, the economic conditions for shipping industry were exacerbated once ships ordered before the crisis were delivered coupled with the crippled world economy. In contrast to the oversupply of the ships during and after the financial crisis, the higher spillovers during the COVID-19 period were owing to the exceptional demand and storage of the oil for the tanker sector. On the other hand, the operations of the dry cargo sector were halted for some time due to the closure of the businesses. But as the businesses started to open the dry cargo sector was unable to meet the peaked demand and shipping industry confronted severe supply glitches.

The higher spillovers can be seen in case of tanker sector from 2009 to 2014. On other hand, oil prices started to recover after 2009, which resulted in an increase in the fuel costs for shipping industry. As a result, slow steaming was a favored method to avoid increasing fuel cost making the delivery time slower for shipping companies. Even though the period from 2013 to 2015 was a moderate period of volatility for tanker sector, but higher spillovers are evident from dry cargo sector. Another contributing factor is the lower consumption of oil than its production during

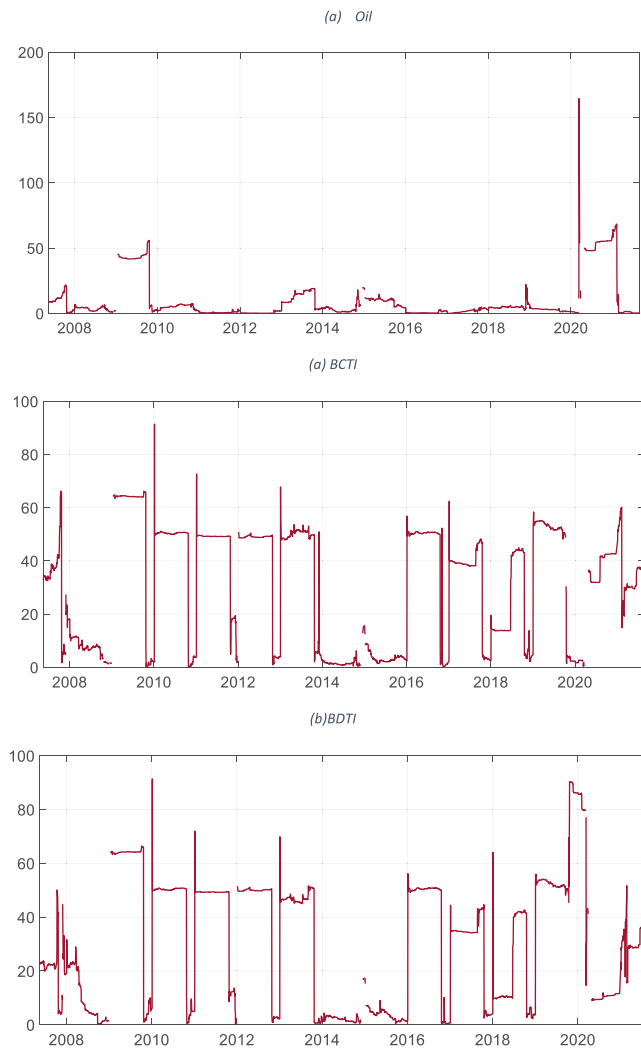


Fig. 3. Volatility spillovers “To Others”-Tanker market.

this period. Whether the volatility in oil prices contributed some spillovers during the period or not can be seen from Fig. 3. The oil contributed major spillovers to dry cargo sector during the period 2014–2016. During this period, oil price depreciated about 70% and it can be categorized as one of the biggest declines since world war-II. The main driver of this drop in oil prices was the supply glut coupled with the lower global demand. However, the drop in oil prices failed to boost the global economy and increase demand. These results indicate that greater spillovers are specifically observed when volatility in oil prices is coupled with lower oil prices. The same can be attributed to the COVID-19 as global financial markets behaved quite similar to the financial crisis. However, the major difference for the shipping industry was the historical low prices for the oil which increased the demand for the tanker sector increased manifold as tankers also serve the purpose of storing oil.

The findings of the towering spillovers following the period of financial crisis are consistent with Awartani and Maghyreh (2013), and Diebold and Yilmaz (2012) who also showed that markets around the world experienced higher volatility spillovers after financial crisis. Shipping industry's performance is highly dependent on the performance of the global economy and its growth is associated with world's demand. Although the observed spillovers started to settle from 2017; however, the trade war after the Trump's administration took office played a key role

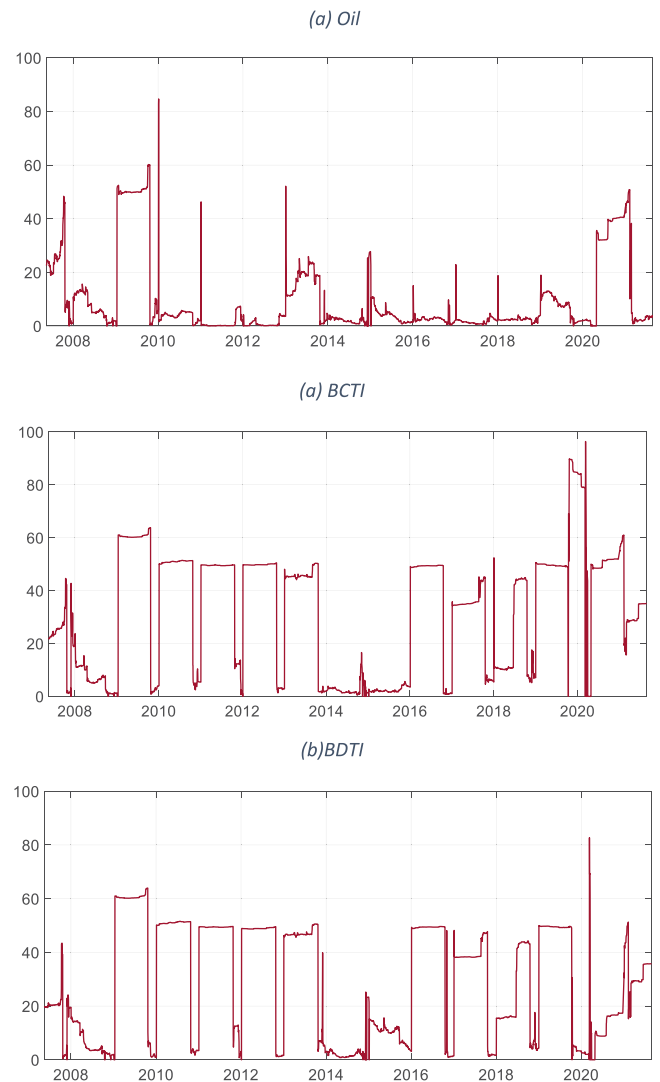


Fig. 4. Volatility spillovers “FROM Others”-Tanker market.

in dwindling the shipping industry again. The higher spillovers starting from the last quarter of 2018 can be attributed to the trade war between China and US which resulted in a decrease of about 2% of the world's maritime trade along with an increase in the volatility in shipping market.

4.1. Directional volatility spillovers

One of the key advantages of using Diebold and Yilmaz's (2012) spillover method is the segregation of spillovers “from” and “to” the assets as it captures the individual dynamics of spillovers emitted or received from each asset under consideration. Fig. 3 shows the directional volatility spillovers emanated from oil and tanker market. Overall, oil transmits lower volatility spillovers than tanker sector. There is a period of large volatility spillovers from oil before financial crisis precisely during the period from 2005 to 2006. Oil prices during this period were highly volatile and touched their peak shortly before the financial crisis. On other hand, spillovers from oil to tanker sector are not distinct after financial crisis except a spike at fall of Lehman Brothers, another noticeable period of volatility spillovers for oil is the period from 2014 to 2016 owing to the volatility in oil market as oil prices were highly volatile during this period. It

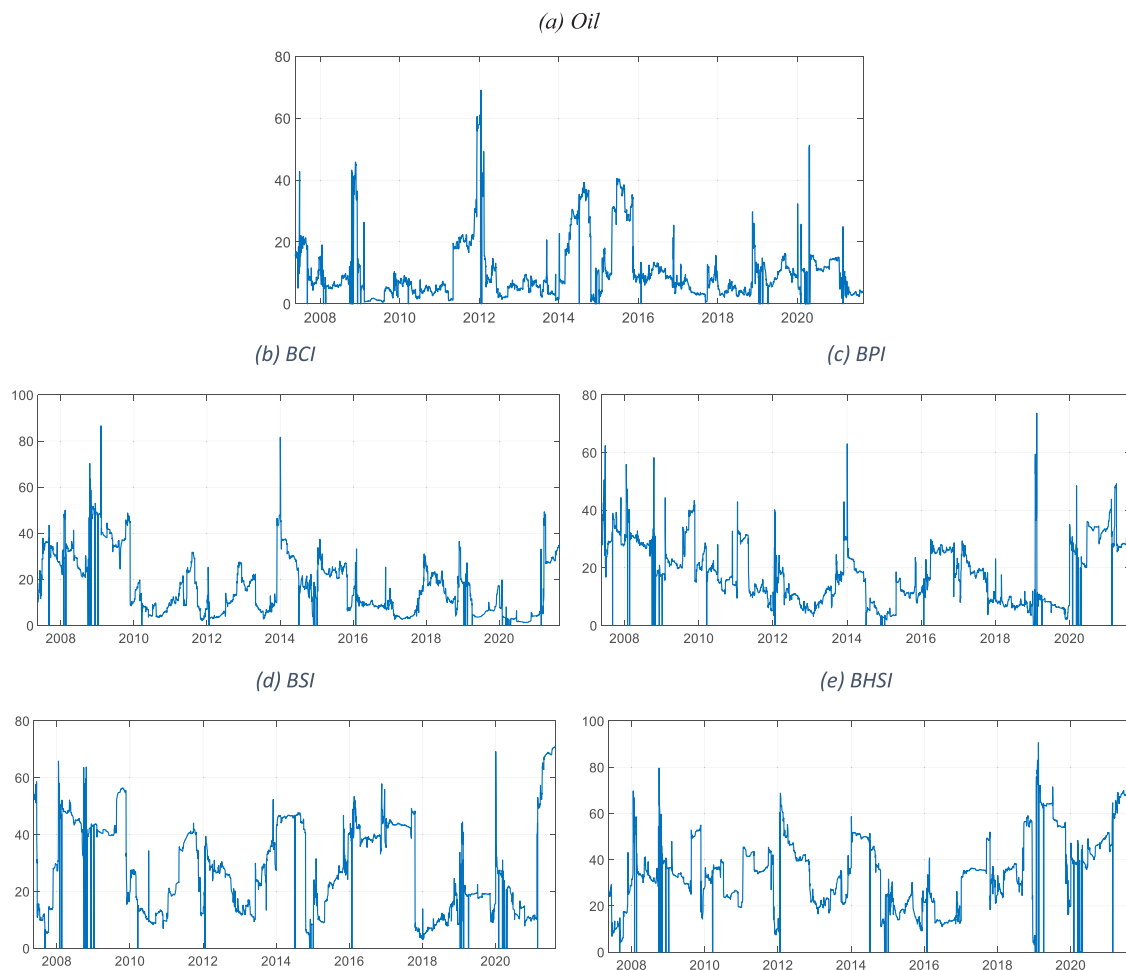


Fig. 5. Volatility Spillover "From Others"- Dry Cargo Market.

confirms the findings of [Chen et al. \(2017\)](#), who analyzed the relationship between tanker market and oil based on Multifractal Detrended Cross-Correlation Analysis (MF-DCCA) and concluded that financial crisis has little influence on the persistent long range correlation between oil and tanker market. The pattern of emitting and receiving noticeable spillovers from the dirty and clean tanker is quite identical to each other. These spillovers for tanker market, as shown in [Figs. 3 and 4](#), also indicate higher interconnectedness among the tanker market. Oil contributed major spillovers during the pandemic period and the same can be seen for the clean tanker market. We can observe a spike at the onset of the COVID-19 period and an equivalent of the spillovers during the financial crisis were found for the period following the pandemic's spike. However, dirty tanker did not contribute much at the start but the spillovers started to increase after some time.

On the receiving end too, the spillovers for volatility in the oil prices remained indistinguishable for most of the period except during the financial crisis. Alternately, tanker market received higher spillovers and dirty tanker sector is observed to be the most vulnerable to spillovers emitted during the period 2014 to 2016 from oil. The directional spillovers show a clear picture of the magnitude, direction and interconnectedness of tanker market, which is mainly responsible for the spillovers and it signifies the role played by the volatility in oil prices. In the case of dry cargo market, the spillovers originating from oil were higher before financial crisis followed by a calm period from 2009 to 2014. In the case of oil, spillovers "from" and "to" the dry cargo sub-sectors were relatively lower in the periods following financial crisis with the exception of high volatile oil

prices' period from the onset of 2014 to 2016. On other hand, we find higher spillovers coupled with the higher number of spikes in comparison to tanker market. It is remarkable that with the increase (decrease) in the ship's size the spillovers to others decreases (increases) for the cargo sector. It is interesting to note that the pattern while receiving spillovers from others is alike for smaller vessels i.e., BSI and BHSI, demonstrating the higher co-movement of the smaller vessels. The same is true in the case of receiving spillovers, except Handymax receives lower spillovers from others as compared to the Supramax. Handymax and Supramax (small size) transmitted higher spillovers from 2014 to 2016 apart from the period following financial crisis. Remarkably, we observe lower spillovers for the tanker sector during this period. On receiving end, BCI and BPI too followed identical pattern signifying the segmentation based on the size in cargo sub-sectors. [Figs. 5 and 6](#) show that oil plays a vital role in transmitting and receiving spillovers during volatile periods and it amounts higher as compared to the tanker market. Which indicates that dry cargo sector also becomes highly vulnerable to the fluctuations in oil prices during volatile periods. All of the dry cargo sectors received higher spillovers during the COVID-19 periods along with the oil. On the other hand, oil did not contribute much to the dry cargo sector during the spread of the pandemic. The major explanation for this is the transportation and storage of oil for which tankers are used. The operations of the dry cargo sector were halted during the period while those were the golden days for the tanker sector as its demand rose multiple times during the period. Due to the employment of the

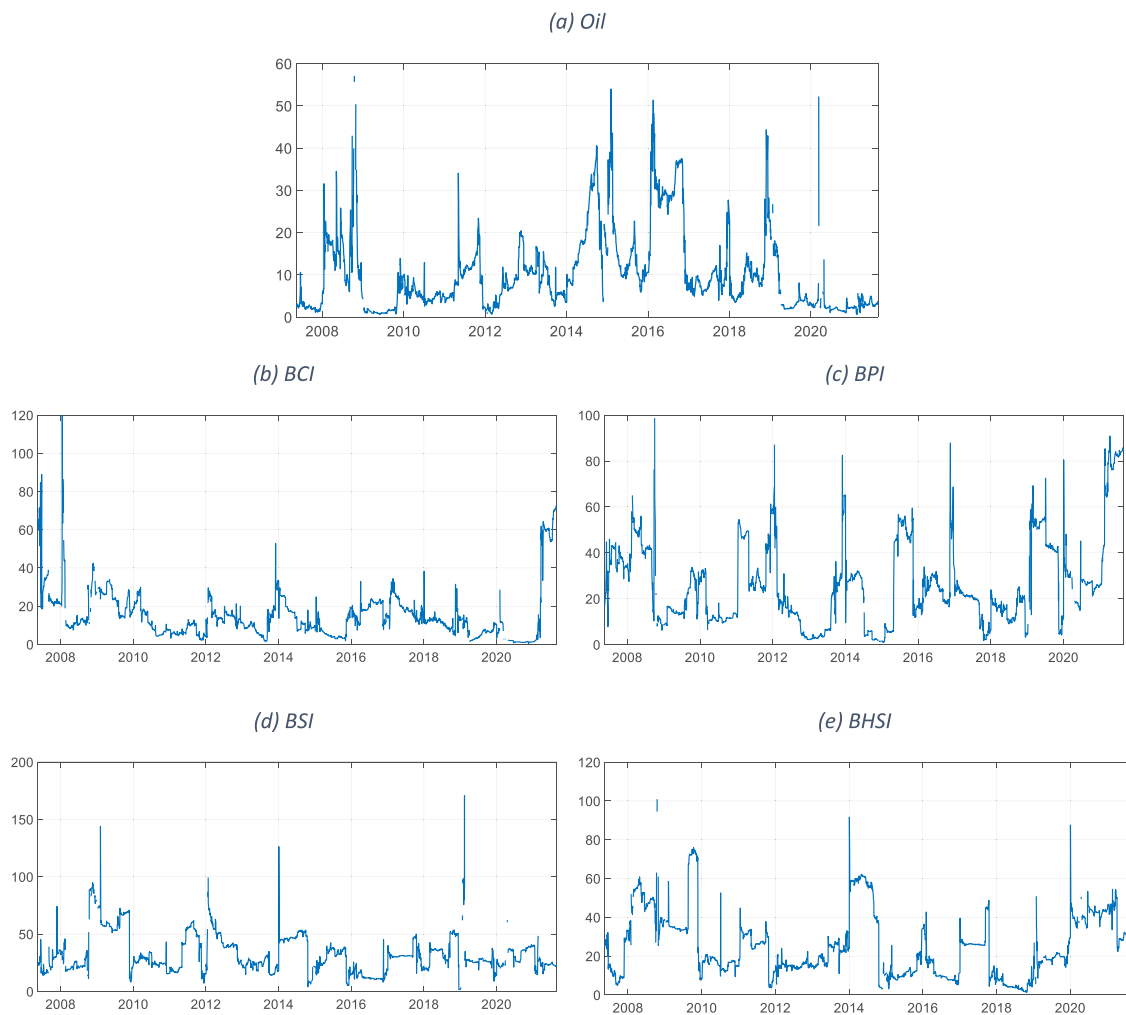


Fig. 6. Volatility Spillover "To Others"- Dry Cargo Market.

tankers for the transportation and storage for the oil, the tanker sector was more affiliated with the oil than the dry cargo sector and the same can be observed in the form of higher spillovers for the tanker sector in comparison to the negligible spillovers for the dry cargo sector. Supramax vessels are found to be responsible for the greater number of shocks to others as can be seen by the higher number of spikes in Fig. 6(d).

The net volatility spillovers can be seen in Fig. 7. We can see pronounced effects before the financial crisis but even after the financial crisis numerous spikes exist in case of clean tanker sector. There are mixed results for dirty tanker market before financial crisis as it was the net receiver of spills, but after the financial crisis it remained net transmitter of volatility spillovers for a short period. It has also received spillovers from the beginning of 2014 to the end of 2018. On the other hand, oil remained a net transmitter of the spillover for the same period; however, the intensity of the spillovers was lower as compared to the dirty tanker. However, the spillovers were intensified after the COVID-19 as is visible in the form of a spike followed by higher net spillovers. The oil remained a net transmitter of the spillovers during this period which gradually diminished by the start of 2021 as oil and other markets stabilized. The chief reason for the surge in spillovers was the sudden drop in oil prices coupled with the shortage of the tankers as those were being used for the storage apart from the transportation of oil. It is further confirmed by the fact that the oil remained a net receiver of spillovers in the cargo sector for the same period.

Regarding dry cargo sector, oil transmitted higher spillovers during and soon after the financial crisis and those amounted to almost 20% for the period. For the remainder period, it had been a net receiver of the spillover with the exception of few years where oil prices were more volatile e.g. the onset of 2016 and 2018. Even during the periods of net transmitting, the intensity of spillovers was not strong enough and was short lived. By looking at the net spillovers we also observe that during the period of higher oil prices, oil mainly remained a net transmitter and received greater spillovers during the period of lower oil prices. Capesize index received spillovers for the most of period among the sectors included in our analysis. However, we find Panamax index to be a net transmitter for the most of the period covered in our analysis. Supramax remained consistently the main emitter of the spillover, while Handymax was on the receiving end with the exception of financial crisis period. Supramax index has transmitted spillovers (about 25%) with spikes reaching to 55%. Interestingly, the net transmitted spillovers were higher during the periods when Capesize (bigger vessels) was a net receiver. Moreover, smaller vessels were net transmitters of the spillovers following the crisis period; however, larger vessels were the net receivers of spillovers. Which indicates that during the more volatile periods, larger vessels markets remain on receiving end while the greater volatility is observed for the smaller vessels market. Handy size market appeared to act an absorbent of the spillovers originated from all the other dry cargo sectors except for the period following the financial crisis. It is pertinent to note

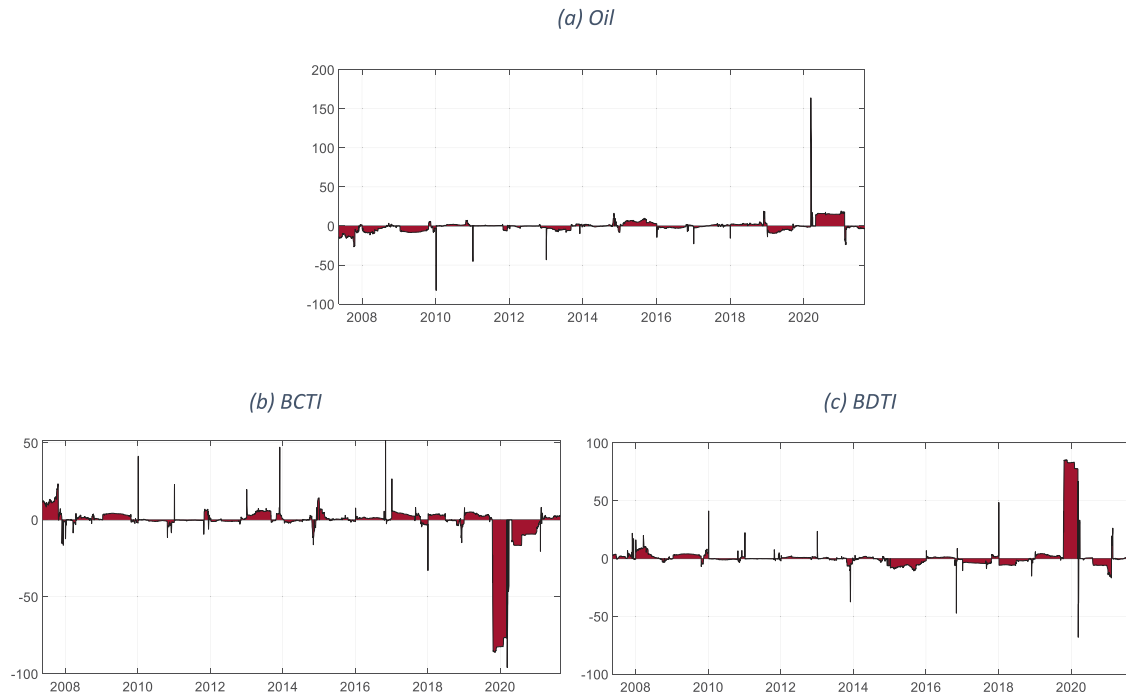


Fig. 7. Net Volatility Spillovers (Tanker Market).

Notes: Rolling-window plots of net spillover index. Note: The dynamic net volatility spillover indices are calculated by subtracting the directional “To” spillovers from the directional “From” spillovers. The positive (negative) value of a spillover for a variable indicates that it is a net contributor (recipient) to the spillover.

that the oil remained a net receiver of the spillovers during the pandemic while it acted as a net transmitter during the same period for the tanker market. During the COVID-19, there was a surge in demand for the tanker sector as oil prices dropped significantly and the economies around the world hunted the advantage of the lower oil prices (see Fig. 8).

4.2. Hedging and portfolio weights

Hedging is a process to avoid future risk from a rise or fall in price. Producers of oil such as oil and gas companies and oil consumer industries (in our case shipping industry) look at hedging differently. Oil production companies usually take a short position in oil futures in order to set the price floor. Therefore, hedging the downside risk. Oil consumer industries on the other hand will normally hedge in case when they foresee an unfavorable event that can lead to a rise in oil price. In order to reduce the risk of investment, we estimate the optimal hedging ratios and optimal portfolio weights for the oil and shipping indices' returns by employing diagonal BEKK model. The estimated conditional covariances matrices from the model are used to calculate optimal hedge ratios and optimal portfolio weights (Kroner and Sultan, 2006), and are shown in Table 4. A long position in asset i can be hedged with a short position in the second asset j and is written as

$$\alpha_{ijt} = \frac{h_{ij,t}}{h_{jj,t}} \quad (10)$$

where α_{ijt} is the hedge ratio between an asset i and asset j ; $h_{ij,t}$ is the conditional covariance of asset i and asset j and $h_{jj,t}$ is the variance of asset j at time t . Following Kroner and Ng (1998), the optimal portfolio weights are computed as

$$w_{ij,t} = \frac{h_{ji,t} - h_{ij,t}}{h_{ii,t} - 2h_{ij,t} + h_{jj,t}} \quad (11)$$

with

$$w_{ij,t} = \begin{cases} 0 & \text{if } w_{ij,t} < 0 \\ w_{ij,t} & \text{if } 0 \leq w_{ij,t} \leq 1 \\ 1 & \text{if } w_{ij,t} > 1 \end{cases}$$

where $w_{ij,t}$ is the weight of asset i in a one dollar portfolio of asset i return and one of the asset j 's return at time t .

The summary statistics for the hedge ratios and portfolio weights for the full period, financial crisis and COVID-19 periods are presented in Tables 4 and 5. For the brevity, we discuss the hedge ratios and portfolio weights for the oil prices and shipping indices only. The average value for the hedge ratio ranges between -0.04 to 0.06 ; however, during the financial crisis period the mean ranges from -0.08 to 0.41 . The positive mean values indicate the long position in the first asset and short position in the second asset and reversal is true for the negative mean values. The results suggest that a long position should be taken for the dry cargo sector with the exception of BHSL; on the other hand, a short position is advised for the tanker sector for the full period. We also notice that hedging becomes expensive during the financial crisis and COVID-19 periods in comparison to the full period. A \$1 long position in the oil can be hedged by shorting 8 cents for BSI which is the most expensive hedging followed by BHSL. A \$1 long position in oil can be hedged for shorting 8 cents for BSI during the full period, however, it increased to shorting 41 cents during the financial crisis and remained at 16 cents during the pandemic period. We observe a bigger change for the hedging ratios for the financial crisis period in comparison to the full period and during pandemic. On the other hand, a short position in oil is favored for the tanker sector for full period and during financial crisis. Moreover, hedging becomes less effective for the dry cargo sector with the exception of BCI during crisis periods such as financial crisis and pandemic. Table 5 shows the summary statistics for the optimal portfolio weights. The average

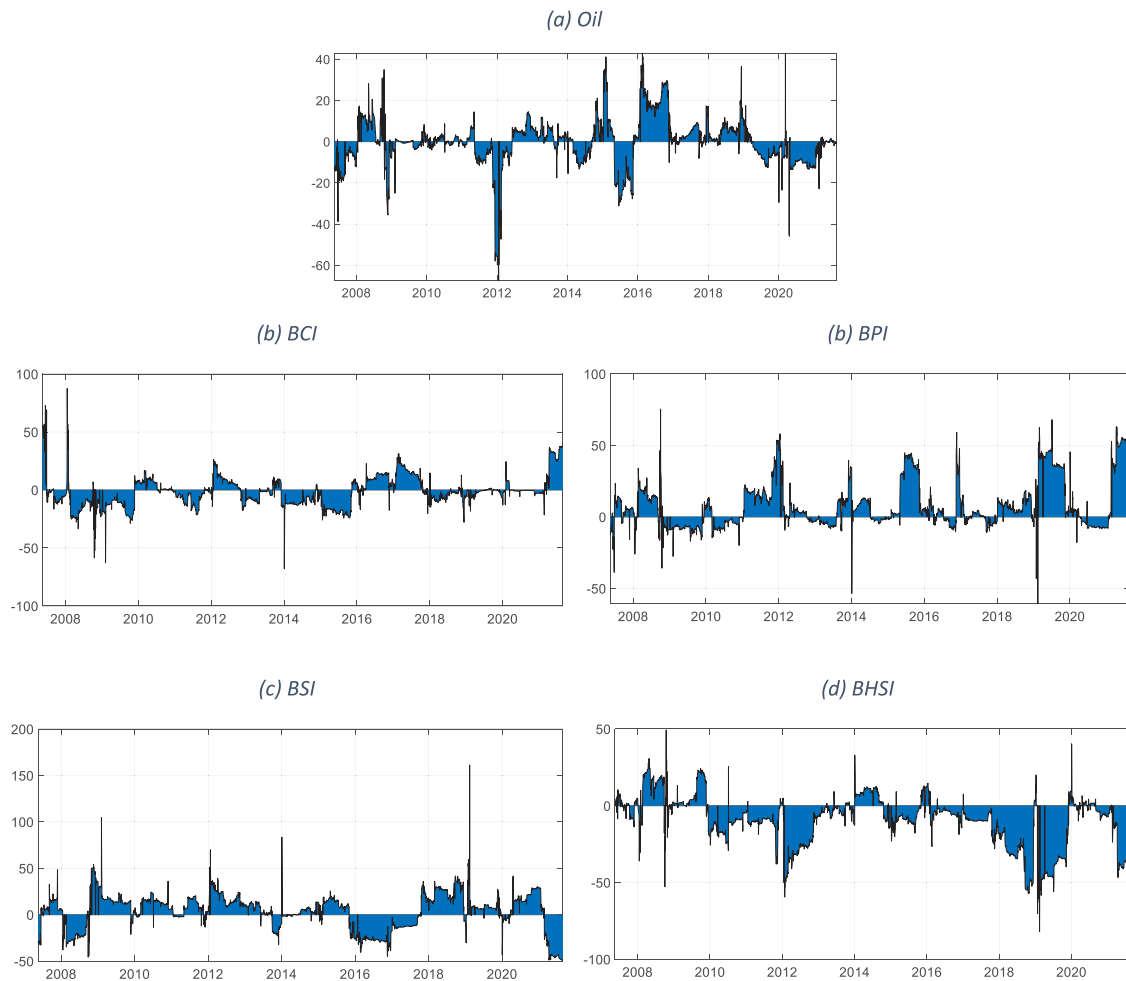


Fig. 8. Net Volatility Spillovers (Dry Cargo).

Notes: Rolling-window plots of net spillover index. Note: The dynamic net volatility spillover indices are calculated by subtracting the directional “To” spillovers from the directional “From” spillovers. The positive (negative) value of a spillover for a variable indicates that it is a net contributor (recipient) to the spillover.

Table 4
Optimal hedge ratios: Summary statistics.

Portfolio pairs	Full period				Financial Crisis (1.12.2007–1.06.2009)				COVID-19 (23.01.2021–31.08.2021)			
	Mean	Stdev	Min	Max	Mean	Stdev	Min	Max	Mean	Stdev	Min	Max
OIL/BCI	0.01	0.1	−0.62	0.75	0.01	0.04	−0.28	0.23	−0.01	0.15	−0.65	1.85
OIL/BPI	0.03	0.18	−1.72	4.89	0.15	0.11	−0.15	0.46	0.09	0.45	−3.46	5.72
OIL/BSI	0.08	0.4	−6.57	7.66	0.41	0.4	−0.25	1.37	0.16	1.33	−12.89	13.56
OIL/BHSI	−0.02	0.38	−4.75	5.84	0.23	0.21	−0.47	1.13	0.04	1.14	−8.58	5.82
BCI/OIL	0.06	1.17	−3.49	46.21	0.01	0.08	−0.83	0.26	0.13	3.44	−26.93	52.72
BPI/OIL	0.03	0.22	−2.32	1.82	0.09	0.08	−0.34	0.39	0.1	0.22	−1.05	1.88
BSI/OIL	0.01	0.13	−0.74	2.85	0.05	0.05	−0.35	0.27	0.02	0.16	−1.64	1.18
BHSI/OIL	0.01	0.09	−0.75	1.11	0.02	0.04	−0.21	0.23	0	0.15	−1.17	1.07
OIL/BCTI	−0.04	0.18	−2.09	0.86	−0.08	0.39	−1.77	1.36	−0.02	0.4	−2.61	2.31
OIL/BDTI	−0.04	0.15	−1.93	0.8	−0.06	0.19	−1.16	0.69	−0.15	0.5	−3.06	2.12
BCTI/OIL	−0.03	0.21	−2.25	3.75	−0.03	0.31	−4.71	0.7	0.02	0.25	−1.37	2.7
BDTI/OIL	−0.04	0.21	−3.67	3.52	−0.1	0.45	−5.06	1.22	−0.02	0.19	−1.48	0.95

weight for Oil/BCTI is 0.43, indicating that for a portfolio of 1\$, 43 cents should be invested in oil and 47 cents be invested in BCTI. With the exception of BCI, our results indicate that more weights should be given to the shipping indices than oil. Moreover, lower weights are assigned to oil market during the financial crisis and COVID-19 period in comparison to the full period. It is also worth noting that the average values do not show the full picture as we observe the higher volatility in the in case of hedging effectiveness and hedging weights specifically during the crisis periods like financial crisis and pandemic.

4.3. Robustness analysis

As a robustness check we use WTI (West Texas Intermediate) as an alternative for the Brent for oil prices. Brent and WTI are widely used in the literature as a benchmark for the crude oil prices. The results obtained by using WTI as a proxy for oil prices does not materially differ from the results computed from using Brent as a proxy for oil and the time varying spillovers remain unchanged as can be seen from Figs. 9 and 10.

In order to check the robustness of the spillover index method, we check the sensitivity of the spillovers through changes in the

Table 5
Optimal portfolio weights: Summary statistics.

Portfolio pairs	Full period				Financial Crisis (1.12.2007–1.06.2009)				COVID-19 (23.01.2021–31.08.2021)			
	Mean	Stdev	Min	Max	Mean	Stdev	Min	Max	Mean	Stdev	Min	Max
OIL/BCI	0.77	0.19	0.07	1	0.49	0.19	0.27	0.94	0.85	0.18	0	1
OIL/BPI	0.49	0.22	0	1	0.45	0.22	0.2	0.96	0.43	0.24	0	1
OIL/BSI	0.22	0.21	0	1	0.22	0.24	0	0.96	0.15	0.17	0	1
OIL/BHSI	0.19	0.16	0	1	0.17	0.19	0	0.93	0.14	0.16	0	1
OIL/BCTI	0.43	0.19	0	1	0.32	0.26	0	0.98	0.38	0.21	0	1
OIL/BDTI	0.48	0.18	0	1	0.55	0.25	0.07	1	0.3	0.18	0	0.99

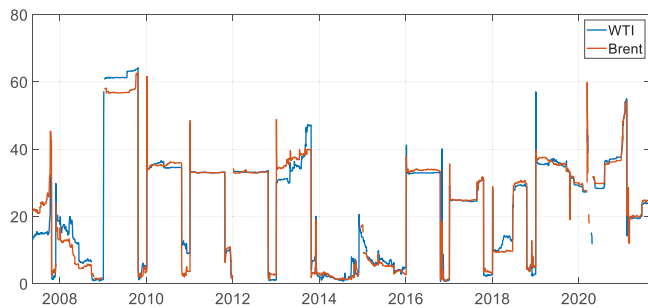


Fig. 9. Total Volatility Spillovers for Tanker Sector (WTI).

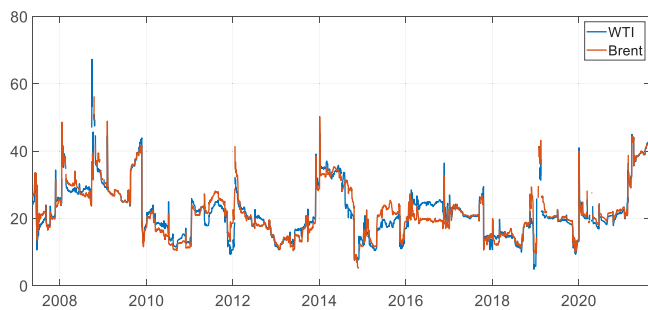


Fig. 10. Total Volatility Spillover for Dry Cargo Sector (WTI).

forecast horizons and selecting different var lengths. As seen from Figs. 11 and 12, the overall pattern of the spillovers remains the same for different forecast horizons and varying lag lengths for both tanker and dry cargo sector. A minimum lag length of 2 is selected while 6 is the maximum lag length we choose. While in order to check the sensitivity for different forecast horizons, a range of 5 to 10 is selected. Although dry cargo market varies about 3% for some of the periods, yet the overall pattern and strength of the relationship holds. On other hand, tanker market strictly follows the same patterns as observed earlier in our results and analysis. Our overall spillovers index holds for the varying degrees of lag lengths and forecast horizon demonstrating that the results are robust to different var lengths and across different forecast horizons.

5. Conclusion and policy implications

Researchers have examined the interconnectedness among different sectors of the marine shipping industry. The marine shipping industry has a strong bond with the oil price as the fuel cost is not only one of the major components of operating cost but tanker sector is also responsible for the transportation of global oil supply. However, missing from the literature is the interconnectedness between the volatility in the oil prices and the marine

shipping industry. In this backdrop, we examined the association of the dry cargo and tanker sector indices with the volatility in oil prices and investigated the magnitude, direction and existence of the dynamic spillovers formulated on the dynamic spillover index methodology suggested by Diebold and Yilmaz (2012). The period extending from May 2006 to August 2021 was gainful since it set forth the dynamics of pre, during and post-crisis period's spillovers along with the COVID-19 period.

We noticed higher spillovers during the financial crisis of 2008 and COVID-19. Dry cargo and tanker market show disparate strength of sensitivities to the oil market. Clean tanker sector was the major contributor and receiver of the spillovers in the case of tanker market specifically during the pandemic, while Handy size (major contributor) and Supramax (major receiver) played significant role in the spillovers in the cargo market sector. Interestingly, the inter-relationship between the different shipping sectors have not changed much over the years and it holds to a larger extent as observed by Drobetz et al. (2012), and Tsouknidis (2016). Moreover, oil contributed higher spillovers in the case of tanker market given the strong dependence of tanker market on oil. Tanker market is also highly integrated among its sectors than with the oil except for the periods of higher volatility in the oil prices. It is noteworthy that the spillovers from the volatility in oil prices increased manifold during the highly volatile periods, and it remained a net receiver of the spillovers from others during the stable periods. The spillovers emitting from oil are conspicuous when oil prices suddenly drop coupled with the higher volatility in the market. This was manifested in the form of supply glut during 2014–2016 for the tanker market. Moreover, time-varying spillovers for oil are congruent with the important economic events such as global financial crisis and during COVID-19. We also observed higher spillovers in the case of smaller vessels (Handy size and Supramax) in comparison to the bigger vessels (Capesize and Panamax) for the dry cargo market. The smaller (larger) the size of the vessels the greater (lower) are the spillovers observed (From and To). Large vessels were on the receiving end when there are higher volatility spillovers emitted from the smaller vessels market, specifically during the periods when overall marine shipping industry is in distress.

The results from our study add in the previous literature by considering the highly relevant and largely ignored implications of the volatility in oil prices for marine shipping industry. Our results from the dynamic relationship between oil and shipping industry can serve as a useful tool for hedging. Investors who hold portfolios of marine shipping firms must consider their exposure to oil prices volatility for tanker and oil market and special care should be taken while diversifying between smaller and larger vessels in the case of dry cargo market. The diversification benefit wanes during highly volatile periods owing to the higher co-movement of the shipping sectors and the same holds for the volatility in oil prices during major economic events.

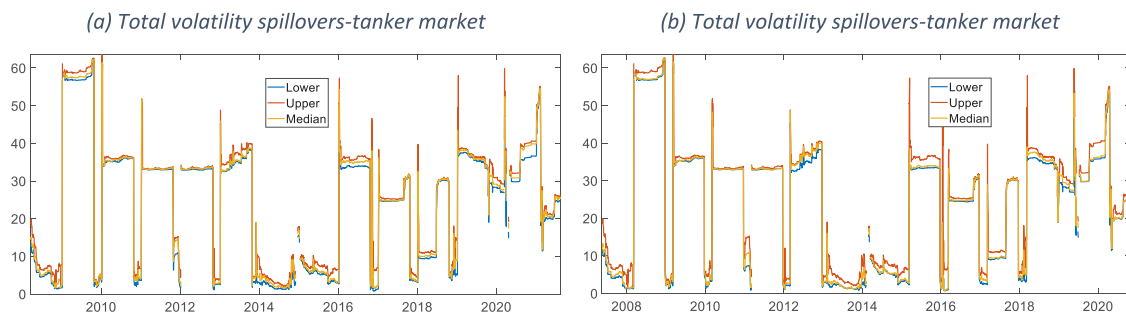


Fig. 11. Robustness of Volatility Spillovers-(Tanker market).

Notes: Panel (a) presents the sensitivity of the estimated spillovers to different var-lengths (minlag = 2, maxlag = 6). Panel (b) represents the sensitivity of the estimated spillovers to the different forecast horizons (minsteps = 5, maxsteps = 10).

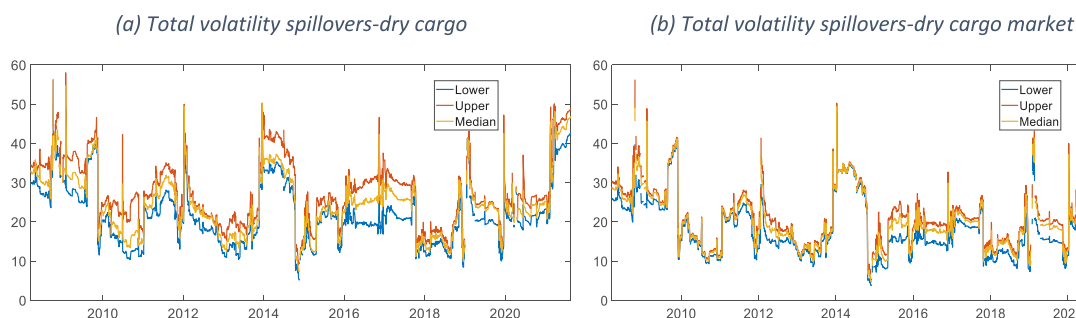


Fig. 12. Robustness of Volatility Spillovers-(Dry cargo market).

Notes: Panel (a) presents the sensitivity of the estimated spillovers to different var-lengths (minlag = 2, maxlag = 6). Panel (b) represents the sensitivity of the estimated spillovers to the different forecast horizons (minsteps = 5, maxsteps = 10).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This research funded by Philosophy and Social Science Laboratory Funding of China Ministry of Education: Evaluation, Simulation and Design of Regional Development Strategies and Policies (ZX2022103). We are thankful to the anonymous referees for their constructive comments which has helped to improve the paper considerably.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.egy.2023.02.025>.

References

- Abadie, L.M., Goicoechea, N., Galarraga, I., 2017. Adapting the shipping sector to stricter emissions regulations: Fuel switching or installing a scrubber. *Transp. Res. D* 57, 237–250. <https://doi.org/10.1016/j.trd.2017.09.017>.
- Aggarwal, R., Akhigbe, A., Mohanty, S.K., 2012. Oil price shocks and transportation firm asset prices. *Energy Econ.* 34 (5), 1370–1379. <https://doi.org/10.1016/j.eneco.2012.05.001>.
- Akinsola, M.O., Odhiambo, N.M., 2020. Asymmetric effect of oil price on economic growth: Panel analysis of low-income oil-importing countries. *Energy Rep.* 6, 1057–1066. <https://doi.org/10.1016/j.egy.2020.04.023>.

- Al-hajji, E., Al-Mulali, U., Solarin, S.A., 2018. Oil price shocks and stock returns nexus for Malaysia: Fresh evidence from nonlinear ARDL test. *Energy Rep.* 4, 624–637. <https://doi.org/10.1016/j.egy.2018.10.002>.
- Alamgir, F., Amin, S. Bin, 2021. The nexus between oil price and stock market: Evidence from South Asia. *Energy Rep.* 7, 693–703. <https://doi.org/10.1016/j.egy.2021.01.027>.
- Anton Mork, K., Olsen, O., Terje Mysen, H., 1994. Macroeconomic responses to oil price increases and decreases in seven OECD countries. *Energy J.* 15 (4), <https://doi.org/10.5547/ISSN0195-6574-EJ-Vol15-No4-2>.
- Awartani, B., Maghyereh, A.I., 2013. Dynamic spillovers between oil and stock markets in the Gulf cooperation council countries. *Energy Econ.* 36, 28–42. <https://doi.org/10.1016/j.eneco.2012.11.024>.
- Aydin, L., Acar, M., 2011. Economic impact of oil price shocks on the Turkish economy in the coming decades: A dynamic CGE analysis. *Energy Policy* 39 (3), 1722–1731. <https://doi.org/10.1016/j.enpol.2010.12.051>.
- Basher, S.A., Haug, A.A., Sadorsky, P., 2012. Oil prices, exchange rates and emerging stock markets. *Energy Econ.* 34 (1), 227–240. <https://doi.org/10.1016/j.eneco.2011.10.005>.
- Bialystocki, N., Konovessis, D., 2016. On the estimation of ship's fuel consumption and speed curve: A statistical approach. *J. Ocean Eng. Sci.* 1 (2), 157–166. <https://doi.org/10.1016/j.joes.2016.02.001>.
- Chen, S., Meersman, H., van de Voorde, E., 2010. Dynamic interrelationships in returns and volatilities between Capesize and Panamax markets. *Marit. Econ. Logist.* 12 (1), 65–90. <https://doi.org/10.1057/mel.2009.19>.
- Chen, F., Miao, Y., Tian, K., Ding, X., Li, T., 2017. Multifractal cross-correlations between crude oil and tanker freight rate. *Physica A: Stat. Mech. Appl.* 474, 344–354. <https://doi.org/10.1016/j.physa.2017.01.069>.
- Choi, K., Hammoudeh, S., 2010. Volatility behavior of oil, industrial commodity and stock markets in a regime-switching environment. *Energy Policy* 38 (8), 4388–4399. <https://doi.org/10.1016/j.enpol.2010.03.067>.
- Chou, M.-T., Chou, T.-Y., Hsu, Y.-R., Lu, C.-P., 2017. Fuel consumption ratio analysis for transiting from various ports and harbours in Asia through the northern sea route. *J. Navig.* 70 (04), 859–869. <https://doi.org/10.1017/S0373463317000078>.
- Cifarelli, G., Paladino, G., 2010. Oil price dynamics and speculation: a multivariate financial approach. *Energy Econ.* <https://doi.org/10.1016/j.eneco.2009.08.014>.
- Cong, R.-G., Wei, Y.-M., Jiao, J.-L., Fan, Y., 2008. Relationships between oil price shocks and stock market: An empirical analysis from China. *Energy Policy* 36 (9), 3544–3553. <https://doi.org/10.1016/j.enpol.2008.06.006>.
- Dai, L., Hu, H., Chen, F., Zheng, J., 2014. Volatility transmission in the dry bulk newbuilding and secondhand markets: An empirical research. *Transp. Lett.* 6 (2), 57–66. <https://doi.org/10.1179/1942787514Y.0000000013>.

- Diaz, E.M., Molero, J.C., Perez de Gracia, F., 2016. Oil price volatility and stock returns in the G7 economies. *Energy Econ.* 54, 417–430. <http://dx.doi.org/10.1016/j.eneco.2016.01.002>.
- Dickey, D.A., Fuller, W.A., 1981. Likelihood ratio statistics for autoregressive time series with a unit root. *Econometrica* 49 (4), 1057. <http://dx.doi.org/10.2307/1912517>.
- Diebold, F.X., Yilmaz, K., 2009. Measuring financial asset return and volatility spillovers, with application to global equity markets. *Econ. J.* 119 (534), 158–171. <http://dx.doi.org/10.1111/j.1468-0297.2008.02208.x>.
- Diebold, F.X., Yilmaz, K., 2012. Better to give than to receive: Predictive directional measurement of volatility spillovers. *Int. J. Forecast.* 28 (1), 57–66. <http://dx.doi.org/10.1016/j.ijforecast.2011.02.006>.
- Drobetz, W., Richter, T., Wambach, M., 2012. Dynamics of time-varying volatility in the dry bulk and tanker freight markets. *Appl. Financial Econ.* 22 (16), 1367–1384. <http://dx.doi.org/10.1080/09603107.2012.657349>.
- Drobetz, W., Schilling, D., Tegtmeier, L., 2010. Common risk factors in the returns of shipping stocks. *Marit. Policy Manag.* 37 (2), 93–120. <http://dx.doi.org/10.1080/03088830903533726>.
- Du, X., Yu, C.L., Hayes, D.J., 2011. Speculation and volatility spillover in the crude oil and agricultural commodity markets: A Bayesian analysis. *Energy Econ.* 33 (3), 497–503. <http://dx.doi.org/10.1016/j.eneco.2010.12.015>.
- El-Sharif, I., Brown, D., Burton, B., Nixon, B., Russell, A., 2005. Evidence on the nature and extent of the relationship between oil prices and equity values in the UK. *Energy Econ.* 27 (6), 819–830. <http://dx.doi.org/10.1016/j.eneco.2005.09.002>.
- Faff, R.W., Brailsford, T.J., 1999. Oil price risk and the Australian stock market. *J. Energy Finance Dev.* 4 (1), 69–87. [http://dx.doi.org/10.1016/S1085-7443\(99\)00005-8](http://dx.doi.org/10.1016/S1085-7443(99)00005-8).
- Fagerholt, K., Laporte, G., Norstad, I., 2010. Reducing fuel emissions by optimizing speed on shipping routes. *J. Oper. Res. Soc.* 61 (3), 523–529. <http://dx.doi.org/10.1057/jors.2009.77>.
- Fasanya, I., Akinbawale, S., 2019. Modelling the return and volatility spillovers of crude oil and food prices in Nigeria. *Energy* 169, 186–205. <http://dx.doi.org/10.1016/j.energy.2018.12.011>.
- Fei, Y., Chen, J., Wan, Z., Shu, Y., Xu, L., Li, H., Bai, Y., Zheng, T., 2020. Crude oil maritime transportation: Market fluctuation characteristics and the impact of critical events. *Energy Rep.* 6, 518–529. <http://dx.doi.org/10.1016/j.egy.2020.02.017>.
- Hamilton, J.D., 1983. Oil and the macroeconomy since world war II. *J. Polit. Econ.* 91 (2), 228–248. <http://dx.doi.org/10.1086/261140>.
- Hedi Aroui, M.E., Khuong Nguyen, D., 2010. Oil prices, stock markets and portfolio investment: Evidence from sector analysis in Europe over the last decade. *Energy Policy* 38 (8), 4528–4539. <http://dx.doi.org/10.1016/j.enpol.2010.04.007>.
- Huang, Y., Li, F., 2022. Research on the time-varying spillover effect of international crude oil price on China's exchange rate. *Energy Rep.* 8, 138–148. <http://dx.doi.org/10.1016/j.egy.2022.03.067>.
- Jones, C.M., Kaul, G., 1996. Oil and the stock markets oil and the stock markets. *J. Finance* 51 (2), 463–491. <http://www.jstor.org/stable/2329368>.
- Junttila, J., Pesonen, J., Raatikainen, J., 2018. Commodity market based hedging against stock market risk in times of financial crisis: The case of crude oil and gold. *J. Int. Financial Mark. Inst. Money* 56, 255–280. <http://dx.doi.org/10.1016/j.intfin.2018.01.002>.
- Kang, S.H., McIver, R., Yoon, S.-M., 2017. Dynamic spillover effects among crude oil, precious metal, and agricultural commodity futures markets. *Energy Econ.* 62, 19–32. <http://dx.doi.org/10.1016/j.eneco.2016.12.011>.
- Kavussanos, M.G., 1997. The dynamics of time-varying volatilities in different size second-hand ship prices of the dry-cargo sector. *Appl. Econ.* 29 (4), 433–443. <http://dx.doi.org/10.1080/000368497326930>.
- Kavussanos, M.G., 2003. Time varying risks among segments of the tanker freight markets. *Marit. Econ. Logist.* 5 (3), 227–250. <http://dx.doi.org/10.1057/palgrave.mel.9100079>.
- Kavussanos, M.G., Marcoulis, S.N., 1997. The stock market perception of industry risk and microeconomic factors: The case of the US water transportation industry versus other transport industries. *Transp. Res. Part E: Logist. Transp. Rev.* 33 (2), 147–158. [http://dx.doi.org/10.1016/S1366-5545\(97\)00015-X](http://dx.doi.org/10.1016/S1366-5545(97)00015-X).
- Kavussanos, M.G., Visvikis, I.D., 2016. The international handbook of shipping finance. *Theory Pract.* 1–430. <http://dx.doi.org/10.1057/978-1-137-46546-7>.
- Kilian, L., Vigfusson, R.J., 2011. Are the responses of the U.S. economy asymmetric in energy price increases and decreases? *Quant. Econ.* 2 (3), 419–453. <http://dx.doi.org/10.3982/QE99>.
- Kocaarslan, B., Soytaş, U., 2019. Asymmetric pass-through between oil prices and the stock prices of clean energy firms: New evidence from a nonlinear analysis. *Energy Rep.* 5, 117–125. <http://dx.doi.org/10.1016/j.egy.2019.01.002>.
- Koop, G., Pesaran, M.H., Potter, S.M., 1996. Impulse response analysis in nonlinear multivariate models. *J. Econometrics* 74 (1), 119–147. [http://dx.doi.org/10.1016/0304-4076\(95\)01753-4](http://dx.doi.org/10.1016/0304-4076(95)01753-4).
- Kristjansson, W.D., Concha, D., 2016. Impact of fuel price fluctuations on airline stock returns. *Appl. Energy* 178, 496–504. <http://dx.doi.org/10.1016/j.apenergy.2016.06.089>.
- Kroner, K.F., Ng, V.K., 1998. Modeling asymmetric comovements of asset returns. *Rev. Financ. Stud.* 11 (4), 817–844. <http://dx.doi.org/10.1093/rfs/11.4.817>.
- Kroner, K.F., Sultan, J., 2006. Time-varying distributions and dynamic hedging with foreign currency futures. *J. Financial Quantit. Anal.* <http://dx.doi.org/10.2307/2331164>.
- Lau, Y.-Y., Ng, A.K.Y., Fu, X., Li, K.X., 2013. Evolution and research trends of container shipping. *Marit. Policy Manag.* 40 (7), 654–674. <http://dx.doi.org/10.1080/03088839.2013.851459>.
- Lee, K., Ni, S., 2002. On the dynamic effects of oil price shocks: A study using industry level data. *J. Monetary Econ.* 49 (4), 823–852. [http://dx.doi.org/10.1016/S0304-3932\(02\)00114-9](http://dx.doi.org/10.1016/S0304-3932(02)00114-9).
- Lee, K., Ni, S., Ratti, R.A., 1995. Oil shocks and the macroeconomy: The role of price variability. *Energy J.* 16 (4). <http://dx.doi.org/10.5547/ISSN0195-6574-EJ-Vol16-No4-2>.
- Li, Y., Huang, J., Gao, W., Zhang, H., 2021. Analyzing the time-frequency connectedness among oil, gold prices and BRICS geopolitical risks. *Resour. Policy* 73 (May), 102134. <http://dx.doi.org/10.1016/j.resourpol.2021.102134>.
- Luo, J., Ji, Q., 2018. High-frequency volatility connectedness between the US crude oil market and China's agricultural commodity markets. *Energy Econ.* 76, 424–438. <http://dx.doi.org/10.1016/j.eneco.2018.10.031>.
- Maitra, D., Chandra, S., Dash, S.R., 2020. Liner shipping industry and oil price volatility: Dynamic connectedness and portfolio diversification. *Transp. Res. Part E: Logist. Transp. Rev.* 138 (May), 101962. <http://dx.doi.org/10.1016/j.tre.2020.101962>.
- Malik, F., Hammoudeh, S., 2007. Shock and volatility transmission in the oil, US and Gulf equity markets. *Int. Rev. Econ. Finance* 16 (3), 357–368. <http://dx.doi.org/10.1016/j.iref.2005.05.005>.
- Mohanty, S., Nandha, M., Habis, E., Juhabi, E., 2014. Oil price risk exposure: The case of the U.S. travel and Leisure industry. *Energy Econ.* 41, 117–124. <http://dx.doi.org/10.1016/j.eneco.2013.09.028>.
- Mokni, K., 2020. Time-varying effect of oil price shocks on the stock market returns: Evidence from oil-importing and oil-exporting countries. *Energy Rep.* 6, 605–619. <http://dx.doi.org/10.1016/j.egy.2020.03.002>.
- Narayan, P.K., Sharma, S.S., 2011. New evidence on oil price and firm returns. *J. Bank. Financ.* 35 (12), 3253–3262. <http://dx.doi.org/10.1016/j.jbankfin.2011.05.010>.
- Nazlioglu, S., Erdem, C., Soytaş, U., 2013. Volatility spillover between oil and agricultural commodity markets. *Energy Econ.* 36, 658–665. <http://dx.doi.org/10.1016/j.eneco.2012.11.009>.
- Park, J., Ratti, R.A., 2008. Oil price shocks and stock markets in the U.S. and 13 European countries. *Energy Econ.* 30 (5), 2587–2608. <http://dx.doi.org/10.1016/j.eneco.2008.04.003>.
- Peng, C., Zhu, H., Guo, Y., Chen, X., 2018. Risk spillover of international crude oil to China's firms: Evidence from granger causality across quantile. *Energy Econ.* 72, 188–199. <http://dx.doi.org/10.1016/j.eneco.2018.04.007>.
- Pesaran, H.H., Shin, Y., 1998. Generalized impulse response analysis in linear multivariate models. *Econom. Lett.* 58 (1), 17–29. [http://dx.doi.org/10.1016/S0165-1765\(97\)00214-0](http://dx.doi.org/10.1016/S0165-1765(97)00214-0).
- Phan, D.H.B., Sharma, S.S., Narayan, P.K., 2015. Oil price and stock returns of consumers and producers of crude oil. *J. Int. Financial Mark. Inst. Money* 34, 245–262. <http://dx.doi.org/10.1016/j.intfin.2014.11.010>.
- Psaraftis, H.N., Kontovas, C.A., 2014. Ship speed optimization: Concepts, models and combined speed-routing scenarios. *Transp. Res. C* 44, 52–69. <http://dx.doi.org/10.1016/j.trc.2014.03.001>.
- Ramos, S.B., Veiga, H., 2011. Risk factors in oil and gas industry returns: International evidence. *Energy Econ.* 33 (3), 525–542. <http://dx.doi.org/10.1016/j.eneco.2010.10.005>.
- Ronen, D., 1982. The effect of oil price on the optimal speed of ships. *J. Oper. Res. Soc.* 33 (11), 1035–1040. <http://dx.doi.org/10.1057/jors.1982.215>.
- Sadorsky, P., 1999. Oil price shocks and stock market activity. *Energy Econ.* 21 (2), 449–469. [http://dx.doi.org/10.1016/S0140-9883\(99\)00020-1](http://dx.doi.org/10.1016/S0140-9883(99)00020-1).
- Sadorsky, P., 2008. Assessing the impact of oil prices on firms of different sizes: Its tough being in the middle. *Energy Policy* 36 (10), 3854–3861. <http://dx.doi.org/10.1016/j.enpol.2008.07.019>.
- Sadorsky, P., 2012. Correlations and volatility spillovers between oil prices and the stock prices of clean energy and technology companies. *Energy Econ.* 34 (1), 248–255. <http://dx.doi.org/10.1016/j.eneco.2011.03.006>.
- Saidu, M.T., Naseem, N.A.M., Law, S.H., Yasmin, B., 2021. Exploring the asymmetric effect of oil price on exchange rate: Evidence from the top six African net oil importers. *Energy Rep.* 7, 8238–8257. <http://dx.doi.org/10.1016/j.egy.2021.07.037>.
- Scholten, B., Yurtsever, C., 2012. Oil price shocks and European industries. *Energy Econ.* 34 (4), 1187–1195. <http://dx.doi.org/10.1016/j.eneco.2011.10.012>.
- Singh, V.K., Nishant, S., Kumar, P., 2018. Dynamic and directional network connectedness of crude oil and currencies: Evidence from implied volatility. *Energy Econ.* 76, 48–63. <http://dx.doi.org/10.1016/j.eneco.2018.09.018>.
- Singhal, S., Ghosh, S., 2016. Returns and volatility linkages between international crude oil price, metal and other stock indices in India: Evidence from VAR-DCC-GARCH models. *Resour. Policy* 50 (C), 276–288. <http://dx.doi.org/10.1016/j.resourpol.2016.10.001>.

- Smith, A., 1776. The wealth of nations. In: Book, Vol. 2, no. 56. <http://dx.doi.org/10.2307/2221259>.
- Soytas, U., Oran, A., 2011. Volatility spillover from world oil spot markets to aggregate and electricity stock index returns in Turkey. *Appl. Energy* 88 (1), 354–360. <http://dx.doi.org/10.1016/j.apenergy.2010.07.018>.
- Soytas, U., Sari, R., Hammoudeh, S., Hacihasanoglu, E., 2009. World oil prices, precious metal prices and macroeconomy in Turkey. *Energy Policy* 37 (12), 5557–5566. <http://dx.doi.org/10.1016/j.enpol.2009.08.020>.
- Stopford, M., 2009. *Maritime Economics*, third ed. Routledge, [http://dx.doi.org/10.1016/S0966-6923\(98\)00021-0](http://dx.doi.org/10.1016/S0966-6923(98)00021-0).
- Sun, X., Haralambides, H., Liu, H., 2019. Dynamic spillover effects among derivative markets in tanker shipping. *Transp. Res. Part E: Logist. Transp. Rev.* 122 (2018), 384–409. <http://dx.doi.org/10.1016/j.tre.2018.12.018>.
- Tiwari, A.K., Aye, G.C., Gupta, R., Gkillas, K., 2020. Gold-oil dependence dynamics and the role of geopolitical risks: Evidence from a Markov-switching time-varying Copula model. *Energy Econ.* 88, 104748. <http://dx.doi.org/10.1016/j.eneco.2020.104748>.
- Tiwari, A.K., Jena, S.K., Mitra, A., Yoon, S.M., 2018. Impact of oil price risk on sectoral equity markets: Implications on portfolio management. *Energy Econ.* 72, 120–134. <http://dx.doi.org/10.1016/j.eneco.2018.03.031>.
- Tiwari, A.K., Sahadudheen, I., 2015. Understanding the nexus between oil and gold. *Resour. Policy* 46, 85–91. <http://dx.doi.org/10.1016/j.resourpol.2015.09.003>.
- Tsouknidis, D.A., 2016. Dynamic volatility spillovers across shipping freight markets. *Transp. Res. Part E: Logist. Transp. Rev.* 91, 90–111. <http://dx.doi.org/10.1016/j.tre.2016.04.001>.
- Tuna, V.E., Tuna, G., Kostak, N., 2021. The effect of oil market shocks on the stock markets: Time-varying asymmetric causal relationship for conventional and Islamic stock markets. *Energy Rep.* 7, 2759–2774. <http://dx.doi.org/10.1016/j.egy.2021.04.057>.
- Unctad, 2017. World investment report 2017 - investment and digital economy. http://unctad.org/en/PublicationChapters/wir2017ch3_en.pdf.
- Veenstra, A.W., Ludema, M.W., 2006. The relationship between design and economic performance of ships. *Marit. Policy Manag.* 33 (2), 159–171. <http://dx.doi.org/10.1080/03088830600612880>.
- Wang, S., Meng, Q., 2012. Sailing speed optimization for container ships in a liner shipping network. *Transp. Res. Part E: Logist. Transp. Rev.* 48 (3), 701–714. <http://dx.doi.org/10.1016/j.tre.2011.12.003>.
- Wang, S., Meng, Q., Liu, Z., 2013. Bunker consumption optimization methods in shipping: A critical review and extensions. *Transp. Res. Part E: Logist. Transp. Rev.* 53 (1), 49–62. <http://dx.doi.org/10.1016/j.tre.2013.02.003>.
- Xu, Y., Han, L., Wan, L., Yin, L., 2019. Dynamic link between oil prices and exchange rates: A non-linear approach. *Energy Econ.* 84, 104488. <http://dx.doi.org/10.1016/j.eneco.2019.104488>.
- Yao, Z., Ng, S.H., Lee, L.H., 2012. A study on bunker fuel management for the shipping liner services. *Comput. Oper. Res.* 39 (5), 1160–1172. <http://dx.doi.org/10.1016/j.cor.2011.07.012>.
- Yun, X., Yoon, S.-M., 2019. Impact of oil price change on airline's stock price and volatility: Evidence from China and South Korea. *Energy Econ.* 78, 668–679. <http://dx.doi.org/10.1016/j.eneco.2018.09.015>.